This is the implementation shortfall formulation of Perold (1988) and differentiates between execution cost and opportunity cost. The execution cost component represents the cost that is incurred in the market during trading. Opportunity cost represents the missed profiting opportunity by not being able to transact all shares at the decision price.

Example: A manager decides to purchase 5000 shares of a stock at $10 but the manager is only able to execute 4000 shares at an average price of $10.50. The stock price at the end of trading is $11.00. And the commission cost is $80, which is reasonable since only 4000 shares traded in this example compared to 5000 shares in the above example. Then implementation shortfall including opportunity cost is:

$$IS = \left(\sum s_j p_j - \sum s_j P_d\right) + \left(S - \sum s_j\right) \cdot (P_n - P_d) + fees$$

It is important to note that in a situation where there are unexecuted shares then the IS formulation does depend upon the ending period stock price P_n but in a situation where all shares do execute then the IS formulation does not depend upon the ending period price P_n.

Furthermore, in situations where we have the average execution price of the order, IS further simplifies to:

$$IS = \sum s_j \cdot \left(P_{avg} - P_d\right) + \left(S - \sum s_j\right) \cdot (P_n - P_d) + fees$$

In our example we have:

$$\begin{aligned} IS &= 4000 \cdot (\$10.50 - \$10.00) + 1000 \cdot (\$11.00 - \$10.00) + \$80 \\ &= \$2000 + \$1000 + \$80 = \$3080 \end{aligned}$$

The breakdown of costs following Perold is: execution cost = $2000, opportunity cost = $1000, and fixed fee = $80.

Expanded Implementation Shortfall (Wayne Wagner)

Our third example shows how to decompose implementation shortfall based on where the costs occur in the investment cycle. It starts with opportunity cost, and further segments the cost into a delay component which represents the missed opportunity of being unable to release the order into the market at the time of the investment decision. "Expanded Implementation Shortfall" is based on the work of Wayne Wagner and is often described as Wagner's Implementation Shortfall. This measurement provides managers with valuable insight into "who" is responsible for which costs. It helps us understand whether the incremental cost was